

NOTE: THIS DRIVER REQUIRES VERSION 7.0 OR GREATER OF THE ASCOM PLATFORM TO BE INSTALLED

ASCOM Driver created specifically for the TTS-160 mount

This driver is written for the TTS-160 Panther telescope. In part, it uses code adapted from the Meade LX200 Classic Driver (<https://github.com/kickitharder/Meade-LX200-Classic--ASCOM-Driver>) as noted in inline comments within the code.

The driving force behind this driver was due to the issues surrounding the other available drivers in use which were general LX200 implementations, particularly felt when ASCOM 6.6 was released near the end of 2022. This driver is intended to be able to be maintained by the Panther community to prevent those issues from occurring in the future (or at least corrected more quickly!).

NOTE: The ASCOM device chooser name will not be updated when new versions are installed. Because of this, a standard name of 'Track The Stars TTS-160' will be used beginning in 354.1.2. As long as the installer was used to install the latest driver, everything will work fine regardless of what the chooser shows. If you want to do a clean install of the driver or just want to fix the chooser name, the procedure below will walk you through this. This is highly recommended to be performed when installing v. 356.0.0 for the first time:

THE FOLLOWING PROCEDURE IS OPTIONAL. YOU WILL LOSE YOUR CURRENT DRIVER SETTINGS!

Note 1: The use of ASCOM Device Hub is merely a convenience to get to the ASCOM Chooser dialog box. If you are more comfortable getting to that dialog box from within another program (NINA, SharpCap, etc...), use that method instead

Note 2: Profile Explorer looks very similar to RegEdit as ASCOM saves the driver information in the registry. Profile Explorer uses internal ASCOM functions to ensure all necessary changes are made to the registry so I highly recommend using this approach as opposed to editing your registry directly

1. If present, uninstall the TTS160 driver using Windows' Add or Remove programs ('ASCOM TTS160 Telescope Driver xxx.x.x' or similar).
2. Open ASCOM 'Profile Explorer'.
3. Expand Telescope Drivers.
4. Select 'ASCOM.TTS160.Telescope', right click, select delete and choose 'OK'. Close out of the window.
5. Open 'ASCOM Device Hub'.
6. Select Tools->Setup Go to the 'Telescope Setup' tab and click 'Choose'. Verify the TTS160 driver is not in the list, choose 'Cancel' and close out of Device Hub.
7. Install the most recent version of the TTS160 ASCOM driver.
8. Repeat step 6, this time verifying that you see 'Track The Stars TTS-160'. Feel free at this time to select 'Properties' and re-enter your driver settings.

THE ABOVE PROCEDURE WAS OPTIONAL AND HAS RESET YOUR DRIVER SETTINGS!

As of v. 354.1.0, the driver is now a local server, no need to go through ASCOM Device Hub or ASCOM Remote. This driver may be selected directly and connected in multiple applications.

As of firmware 355C, the mount supports moving at all rates from 0 to 3.5 deg/sec on both axes. Driver version 355.0.0 and later will automatically detect if the mount firmware version supports this feature. If it detects an earlier firmware version, it will restrict the available speeds to the previous capability. In that case, the slew rate has 3 speed options (1.4, 2.2, 3 deg/sec) as well as a much slower speed, the actual value of which is dependent on the guide rate setting of the hand pad. All client programs such as NINA or Sky Safari will see and be able to use whichever speeds are available based on the detected firmware version.

In addition to the available slewing speeds, version 355 implements several other new features and enhanced commands. The driver will use the full capability available to it while still being able to work with older firmware versions. If running previous driver versions with the 355 firmware, some enhancements will be able to be used (align on sync, for example), but others such as the custom slew speeds, will not be available. In general, best practice is to match the driver and firmware major versions (355, in this case).

Another source of potentially strange behavior is if you go off of the operations screen in the handpad while the driver is trying to communicate with the mount. Currently, remote communications from the handpad generally only happen in the operations screen. If something is not working right, first step should be to completely disconnect (from all programs, even going so far as to using task manager to verify the ASCOM TTS160 driver process is stopped) and then reconnect.

Lastly, be aware that the mount will not resume tracking if a slew is canceled (it will resume tracking when a slew is completed).

As of 356, the mount will support Ra and Dec rates to enable tracking objects that slowly move in Ra and Dec.

The source code for the driver can be found at: <https://github.com/rwsmythe/TTS160-Server>

Enjoy, and let me know if you have any issues.

--Reid

Setup Window Clarifications:

Site altitude is used to report site altitude as needed. If you change site location in the mount, you may also want to change the site altitude in the driver. Please note that programs can NOT update the site location (lat/long) stored in the handpad, altitude CAN be updated.

Slew settling time is essentially a wait period before the mount reports that a slew is complete. There is typically a short settling time after the mount starts tracking before it has completely stopped. This may be useful to set to a larger value to account for rOTAtor motion if you don't want a slew to be considered complete until after the rOTAtor is finished moving. Limited to integer seconds based on ASCOM compliance.

Recorded site lat/long are read-only fields that show what the handpad location was set to the last time the driver read the site location. If it has changed between readings then you will see the old location until you have re-connected the mount and that location is refreshed. The displayed values are for awareness only and do not impact driver or mount operations.

Sync Mount Time to Computer on Connect is self explanatory. There may be situations where you do not wish for this to occur. If that is the case, simply uncheck the box. Note that this syncs both time and date and is based off of the time zone correction on the handpad being correct. This should prevent, for example, NINA warnings about time mismatch and it should improve goto and tracking in general, assuming your computer has accurate time.

Park Location Provides information on the handpad park information from the last time the mount was connected. If 'Update Mount' is selected, the settings entered here will be pushed to the mount when the driver is connected.

Align on Sync sets the mount in to the mode equivalent to 'Align on PC' from the Alignment menu. In this mode, the mount will align itself to the first 1-3 synchronize commands sent (selectable in the driver setup screen). This all allow you to power on the mount, connect the driver, and immediately begin the alignment process with no further action required. On completion, the mount will be aligned and ready to operate. The closer you have the mount pointing south at 0 degrees elevation, the more accurate the initial alignment slews will be. The first alignment star assumes the mount is level which may impact the accuracy of the slew to the second star. Use of a plate solve and sync routine mitigates those potential issues.